

Yunong (Nathan) Shen

[Google Scholar](#)

EDUCATION

Sep. 2021 - Jun. 2026 (expected)	Peking University Doctor of Philosophy in Materials Science & Engineering Capital Medical University Cooperative research in Neuroscience	Beijing, China
Sep. 2023 - Jun. 2026	University of Science & Technology Beijing Bachelor in Materials Science & Engineering	Beijing, China
Sep. 2017 - Jun. 2021		Beijing, China

RESEARCH INTEREST

Additive manufacturing Regenerative Medicine Medical Devices

RESEARCH EXPERIENCE

Jun. 2023 - Jun. 2026 <i>Ph.D. Phase</i>	China-America Institute of Neuroscience - Developed biodegradable self-expanding stent for the treatment of ischemic stroke - Developed biodegradable 3D coil for intracranial aneurysm embolization - Participated in the development of neuroprotective stent - Participated in the development and verification of Nitinol stent retriever with clinical applications in 80 cases
Sep. 2021 - Jun. 2026 <i>Ph.D. Phase</i>	Laboratory of Biomedical Materials and Devices - Developed novel molybdenum based alloys for biodegradable cerebrovascular stenting & coiling, which helped the lab to secure US\$360,000 funding from NSFC. - Explored corrosion mechanism and interface reactions of molybdenum based metals - Explored high cycle corrosion fatigue behavior of molybdenum based metals - Laser Powder Bed Fusion (L-PBF) biodegradable metal porous scaffolds used for the treatment of large bony defects
Jun. 2019 - Sep. 2021 <i>Undergraduate Phase</i>	- Analyzed the effect of Severe Plastic Deformation method (Equal Channel Angular Pressing, ECAP) on comprehensive in vitro properties and in vivo osteogenesis of biodegradable magnesium alloys
Dec. 2020 – Jun. 2021 <i>Undergraduate Phase</i>	Corrosion Control Systems Engineering Institute - Fabricated Selective Laser Melting (SLM) nickel-based superalloy Inconel 718, and analyzed its microstructure, mechanical properties and corrosion behavior under different heat treatment regimes

*Peer-Reviewed Research Papers***FIRST AUTHOR & CO-FIRST AUTHOR**

- [1] **Yunong Shen**⁺, Yiming Huang⁺, Yang Zhang⁺, Chunhao Yu, Hanqi Liu, Dong Bian, Di Wu, Wu Wang, Xunming Ji*, Yufeng Zheng*, Miaowen Jiang*, Ming Li*, "Ultra-fine Molybdenum Wire Braided Neuro-interventional Implants: Bridging Biodegradability and Neurovascular Safety for Stroke Treatment." *Advanced Science* **Accept**, [adv.202511466](#)
- [2] Yang Zhang⁺, **Yunong Shen**⁺, Miaowen Jiang⁺, et al. "Gene-based safety evaluation of molybdenum-containing biomaterials for cardiovascular and cerebrovascular diseases." *Genes & Diseases* (2025): 101516.
- [3] Chaixin Wang⁺, Yutian Luo⁺, **Yunong Shen**⁺, et al. "Multifunctional magnesium alloy scaffolds for osteomyelitis therapy: Integrating corrosion resistance, osteogenic activity, and antibacterial effects." *Chemical Engineering Journal* 505 (2025): 158712.
- [4] **Yunong Shen**, Yan Cheng, Yufeng Zheng*, "Insight into the biodegradation behavior of pure molybdenum under simulated inflammatory condition." *Corrosion Science* 228 (2024): 111816.
- [5] Xiaofeng Cao⁺, Guanjin Chen⁺, **Yunong Shen**⁺, Yiming Huang, Chunhao Yu, Aline M. Thomas*, Xiaofei Han*, "Development and translation of biodegradable metal stents: from heart to brain." *Journal of Materials Science & Technology* **Under review**, [J-MST-D-25-04612](#)
- [6] **Yunong Shen**⁺, Yiming Huang⁺, Miaowen Jiang⁺, Yang Zhang, Hongkang Zhang, Chunhao Yu, Hanqi Liu, Guanjin Chen, Xiaofei Han, Xunming Ji*, Ming Li*, Yufeng Zheng*, "Molybdenum ϕ 50 μ m Wire Braided Degradable Stents with Self-Expanding and Radiopaque Features for Intracranial Atherosclerotic Stenosis" **In preparation**
- [7] **Yunong Shen**⁺, Chaojie Tang⁺, Miaowen Jiang⁺, Yang Zhang, Yiming Huang, Hongkang Zhang, Chunhao Yu, Hanqi Liu, Guanjin Chen, Xiaofei Han, Yufeng Zheng*, Xunming Ji*, Ming Li*, Wu Wang*, "Biodegradable 3D detachable coil for treating cerebrovascular aneurysms" **In preparation**
- [8] **Yunong Shen**, Dafu Chen*, Yoji Mine, Kazuki Takashima, Miaowen Jiang, Ming Li*, Yufeng Zheng*, "Fatigue and corrosion fatigue behaviors of biodegradable Mo and Mo-La alloys in air and Hanks' solutions" **In preparation**

CO-AUTHOR

- [1] Miaowen Jiang⁺; Chunhao Yu⁺; Yiming Huang⁺; Xing Zhao; Shiyi Xu; Hongkang Zhang; **Yunong Shen**; Xiaofei Han; Duo Chen; Kun Wang*; Ming Li*, "Hydrodynamic-derived centrifugal blood pump design: from surface to structure." *Bioactive Materials* 54 (2025): 730-743.
- [2] Hongkang Zhang⁺, Yang Zhang⁺, Sheng Lili⁺, Xiaofeng Cao, Chuanjie Wu, Baoying Song, **Yunong Shen**, Zikai Xu, Ge Song, Hao Sun, Qing Liu, Xunming Ji*, Miaowen Jiang*, Ming Li*, Yufeng Zheng*, "Mechanically robust neuroprotective stent by sequential Mg ions release for ischemic stroke therapy." *Nature communications* 16.1 (2025): 6557.
- [3] Yang Zhang⁺, Miaowen Jiang⁺, Ming Wei⁺, Chuanjie Wu⁺, Yiming Huang, Baoying Song, Yi Xu, Hongkang Zhang, **Yunong Shen**, Di Wu*, Yufeng Zheng*, Ming Li*, Xunming Ji*. "MgSO₄ as a novel hypothermia infusion solution promotes ischemic stroke recovery through Ca²⁺ regulation of neurovascular units." *Theranostics* 15.5 (2025): 1896.
- [4] Shaokang Du, **Yunong Shen**, Yufeng Zheng, Yan Cheng, Xiaoxue Xu*, Dafu Chen*, Dandan Xia*. "Systematic in vitro and in vivo study on biodegradable binary Zn-0.2 at% Rare Earth alloys (Zn-RE: Sc, Y, La – Nd, Sm – Lu)." *Bioactive Materials* 24 (2023): 507-523.
- [5] Dandan Xia, Zhaojun Jia, **Yunong Shen**, Yufeng Zheng*, Yan Cheng, Pan Xiong*, Shaokang Guan, Yuqian Xu, Fan Yang, Yunsong Liu*, Yongsheng Zhou*. "pH Stimuli-Responsive, Rapidly Self-healable Coatings Enhanced the Corrosion Resistance and Osteogenic Differentiation of Mg-1Ca Osteoimplant." *Small* 18.36 (2022): 2106056.
- [6] Yu Qin, Aobo Liu, Hui Guo, **Yunong Shen**, Peng Wen*, Hong Lin, Dandan Xia*, Maximilian Voshage, Yun Tian, Yufeng Zheng*. "Additive manufacturing of Zn-Mg alloy porous scaffolds with enhanced osseointegration: In vitro and in vivo studies." *Acta Biomaterialia* 145 (2022): 403-415.
- [7] Yufeng Zheng*, Dandan Xia, **Yunong Shen**, Yunsong Liu, Yuqian Xu, Peng Wen, Yun Tian, Yuxiao Lai. "Additively manufactured biodegradable metal implants." *Acta Metall Sin* 57.11 (2021): 1499-1520.

- [8] Wenting Li, **Yunong Shen**, Jie Shen, Danni Shen, Xiao Liu, Yufeng Zheng*, Kelvin W.K. Yeung*, Shaokang Guan*, Olga B. Kulyasova, R.Z.Valiev*, "In vitro and in vivo studies on pure Mg, Mg – 1Ca and Mg – 2Sr alloys processed by equal channel angular pressing." *Nano Materials Science* 2.1 (2020): 96-108.
- [9] Yufeng Zheng*, **Yunong Shen**, "Opportunities and challenges of molybdenum as a biodegradable metal." *Acta Metall Sin* **Invited review, under review**
- [10] Chunhao Yu, Hongkang Zhang, **Yunong Shen**, Yiming Huang, Guanjun Chen, Hanqi Liu, Xiaofei Han, Yufeng Zheng*, Xunming Ji*, Miaowen Jiang*, Ming Li*, " Contrastive Analysis of Degradation Behavior for WE43 Magnesium Alloy Stents in Cerebrovascular versus Coronary Hemodynamic Environments " *Journal of Magnesium and Alloys* **Under review, JMA-D-25-00719**
- [11] Liyuan Zhong⁺, Miaowen Jiang⁺, Yumin Luo*, Hanqi Liu, Chunhao Yu, Yiming Huang, **Yunong Shen**, Guanjun Chen, Yufeng Zheng, Jiaxuan Xie, Hao Sun, Qing Liu, Zhi Liu, Di Wu, Xunming Ji*, Yuyou Huang*, Ming Li*, "Lymphatic-Immune Microenvironment Engineering via Rapamycin-Eluting Nanofibers for Meningeal Repair and Neuroprotection in TBI Rats. " *Advanced Science* **Under review, advs.202509802**
- [12] Chunhao Yu⁺, Bixiao Chen⁺, Ming Li*, Miaowen Jiang, Zhe Li, Yang Zhang, **Yunong Shen**, Guanjun Chen, Yufeng Zheng*, Renjie Chai*, Xunming Ji* , "Research hotspots and frontier trends of cerebrovascular stent therapy from 2012 to 2024: A bibliometric analysis. " *Journal of Integrative Neuroscience*, **Under review, JIN39592**
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TECHNOLOGICAL INNOVATIONS

P.R.China Patents

- [1] A method for preparing Mo-Mn alloy and its application for medical devices **Under Examination, 202310015962.3**
- [2] Biodegradable 3D detachable coils for aneurysm embolization **In preparation**
- [3] Biodegradable braided stent **In preparation**
- [4] A temporal-controlled-release multi-functional intracranial stent and its preparation method **In preparation**

SELECTED AWARD & RECOGNITION

- **China Optics Valley scholarship**, top 5%, Peking University (2025)
- **Award for Research excellence**, top 5%, Peking University (2025)
- **Red Avenue New Materials scholarship**, top 5%, Peking University (2024)
- **Award for Research excellence**, top 5%, Peking University (2024)
- **Innovation Elite Training Program by Haier INCAIER**, top 5%, Capital Medical University (2024)
- **Outstanding Graduates of Beijing**, top 5% undergraduates, Ministry of Education of Beijing (2021)
- **National Scholarship**, Highest award for undergraduates in China given to top 3% undergraduates, Ministry of Education of the P.R. China (2020)

RESEARCH SKILLS

Characterization & Methods

Characterization: SEM&EDS, Shape Measurement Laser Microscopy, Micro-CT, X-ray photoelectron spectroscopy, X-Ray Diffractometer, Inductively Coupled Optical Atomic Emission Spectroscopy, Fourier Transform infrared spectroscopy

Mechanical testing: Instron universal testing machine, Instron dynamic and fatigue test system

Corrosion testing: Electrochemical impedance spectroscopy

Biocompatibility testing: Cell viability test, Blood Compatibility Test

SERVICE & ACTIVITIES

Aug. 2018 - Sep. 2018

Volunteer Teaching Group, University of Science & Technology Beijing, *Member*

- Co-organized a two-week-long primary school summer camp for over two hundreds of students in Zhang-ShanYing town
- Gave lectures related to nature and humanities
- Collected 258 books for the students
- Led an 10-member team to provide consulting (one-on-one, face-to-face) for each of the 106 students family

REFERENCES

Prof. Yufeng Zheng (Ph.D. advisor, Chair Professor of Biomaterials)
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